

5G Network Slicing Environment Using MATLAB

Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives

Objective – 1: Analyze the Number of Network Slices

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB.

Observe how different slices can be configured to support diverse service requirements and applications.

Objective – 2: Compare Slice Utilization (%)

Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms)

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency

values to understand the Quality of Service (QoS) requirements of various 5G applications.

Software Required

- MATLAB

Theory

Network slicing is a key feature of 5G that enables a single physical network infrastructure to be

partitioned into multiple logical networks called slices. Each slice is optimized for a specific service category, such as enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), or massive Machine-Type Communications (mMTC). By dynamically allocating network resources, network slicing improves flexibility, resource utilization, and Quality of Service (QoS) while supporting diverse application requirements.

Objective – 1

Analyze the Number of Network Slices

Generate and visualize multiple network slices using MATLAB. Observe how a 5G network can

be logically partitioned into independent slices for different communication services.

MATLAB Code:

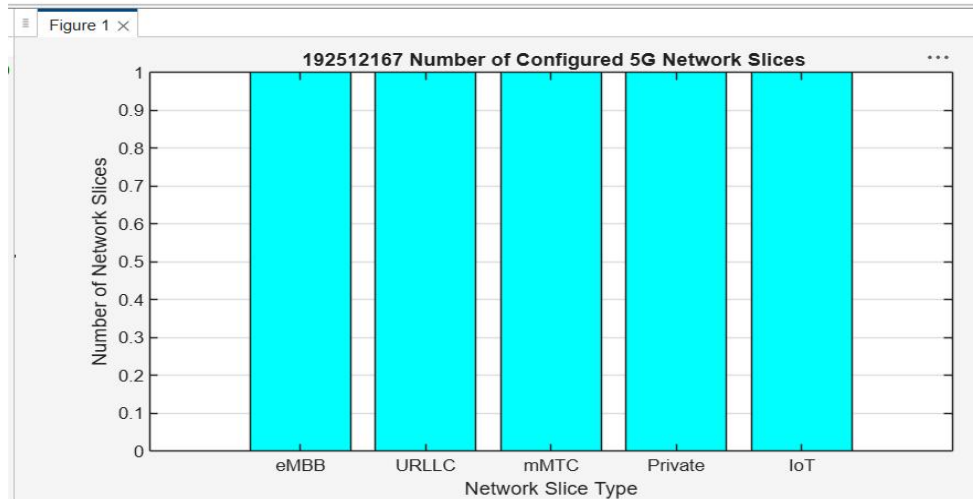
```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1]; % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

```

exp10.m x +
/MATLAB Drive/exp10.m
1  clc;
2  clear;
3  close all;
4  slices = {'eMBB','URLLC','mMTC','Private','IoT'};
5
6  count = [1 1 1 1 1]; % One instance of each network slice
7  figure
8  bar(count,'c')
9
10 grid on
11 set(gca,'XTickLabel',slices)
12 xlabel('Network Slice Type')
13 ylabel('Number of Network Slices')
14 title('192512167 Number of Configured 5G Network Slices')

```

Expected Output



- Bar graph showing the number of configured network slices.

Objective – 2

Compare Slice Utilization (%)

Analyze the utilization percentage of different network slices using MATLAB. Compare the resource usage of each slice to evaluate network efficiency.

MATLAB Code

```

clc;
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];
figure;
bar(slices, utilization);
grid on;
xlabel('Network Slice Type');
ylabel('Slice Utilization (%)');
title('Slice Utilization in 5G Network Slicing');
ylim([0 100]);

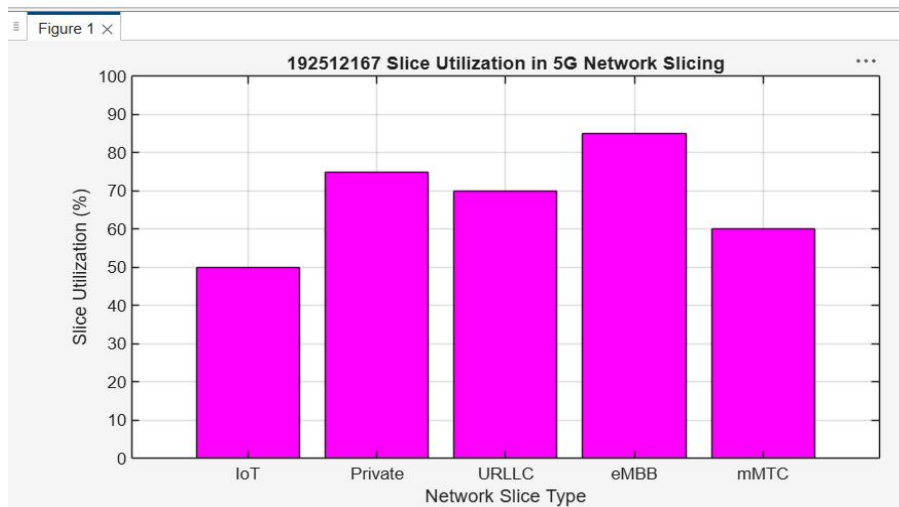
```

```

exp10.m x exp101.m x
/MATLAB Drive/exp101.m
1  clc;
2  clear;
3  close all;
4  slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
5  utilization = [85 70 60 75 50];
6
7  figure;
8  bar(slices, utilization,"magenta");
9  grid on;
10 xlabel('Network Slice Type');
11 ylabel('Slice Utilization (%)');
12 title('192512167 Slice Utilization in 5G Network Slicing');
13 ylim([0 100]);

```

Expected Output



- Bar graph showing slice utilization (%).

Objective – 3

Analyze End-to-End Latency (ms)

Compare the end-to-end latency of different network slices using MATLAB. Observe how latency varies depending on the service requirements of each slice.

MATLAB Code:

```

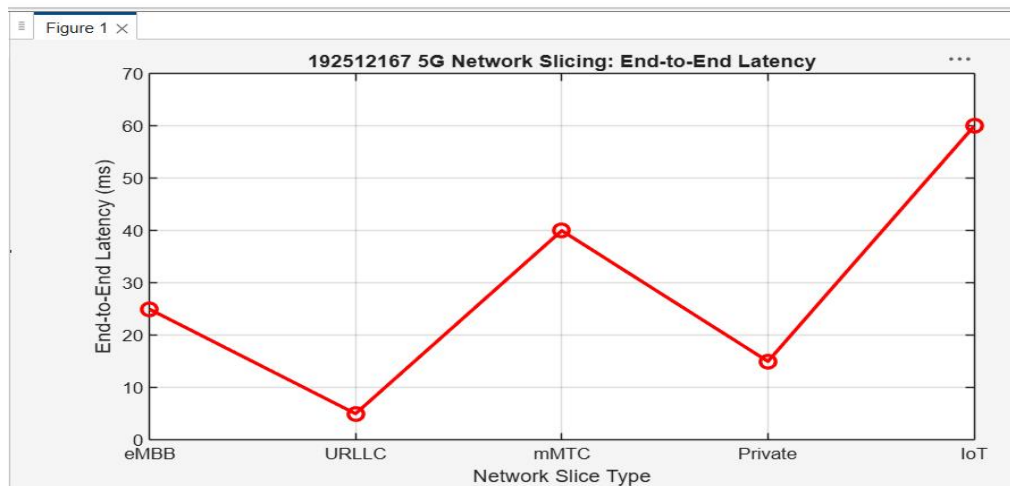
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);

```

Expected Output

```
exp10.m x exp101.m x exp103.m x +
/MATLAB Drive/exp103.m
1  clc;
2  clear;
3  close all;
4  slices = [1 2 3 4 5];
5  latency = [25 5 40 15 60];
6  figure;
7  plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8,'Color','r');
8  grid on;
9
10  xticks(slices);
11  xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
12  xlabel('Network Slice Type');
13  ylabel('End-to-End Latency (ms)');
14  title('192512167 5G Network Slicing: End-to-End Latency');
15  ylim([0 70]);
```

- Line graph showing end-to-end latency for different network slices.



Result

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.